Where AR(2) pooling moves the forecast scores

Main findings of the block-1 positive control, their robustness
to the ridge and recovery issues, and the resulting thesis language

Technical note

31 July 2026

Abstract

The research question is under which simulated dependence conditions the AR(2)-structured skew-t model separates from its i.i.d.-in-time counterpart in Brier score and CRPS. The block-1 answer: separation exists only under near-unit-root dependence AR(2)$(0.15, 0.80)$ and only in the CRPS (-31 to -35%, paired one-sided $p = .039/.012$ under two independent protection schemes); the Brier score at q_{95} never reaches significance anywhere, and the strong-but-short-memory condition AR(2)$(0.80, -0.35)$ shows no detectable difference in either score. These conclusions are invariant to the reflected-ridge pathology (once protected by either remedy) and to every parameter-recovery defect found, which justifies handling both by thesis reminder text rather than by further model changes.

1 Design and test

Settings 4 (AR(2)$(0.80, -0.35)$, memory horizon $h^* = 6$ days) and 5 (AR(2)$(0.15, 0.80)$, $h^* = 109$); datasets 1–10; block 1 (train $t = 1..50$, forecast $t = 51..65$); methods i.i.d. and AR(2), both skew-t, $K = 1$. Per cell and lead h,

$$\mathrm{CRPS}_h = \frac{1}{n_s}\sum_s \mathrm{crps}\big(F_{sh},\, y_{sh}\big), \qquad \mathrm{BS}_h = \frac{1}{n_s}\sum_s \big(\mathbf{1}\{y_{sh} > q_{95}\} - \hat{p}_{sh}\big)^2, \tag{1}$$

with F_{sh} the posterior predictive sample, q_{95} the validation block's own 95% quantile, and $\hat{p}_{sh}$ the posterior predictive exceedance probability. Primary summaries are means over leads 1–5; the test is the paired one-sided Wilcoxon over datasets, $H_1\colon \mathrm{AR}(2) < \mathrm{i.i.d.}$

2 Main result

Three statements summarize Table 1.

1. **CRPS separates, and only under near-unit-root dependence.** The gain under AR(2)$(0.15, 0.80)$ is large, -31 to -35%, and significant under both protection schemes. Under AR(2)$(0.80, -0.35)$ the point estimate is small and unstable across clean-set definitions (the HN row's $p = .049$ excludes dataset 6, the hardest dataset; it should not be read as a discovery).
2. **The Brier score at q_{95} separates nowhere.** Point estimates favor AR(2) under near-unit-root dependence (-11 to -16%) but p never falls below 0.27. This matches the earlier

Table 1: Paired comparison, means over clean datasets, leads 1–5. "guarded" = assertion-guarded study, $\lambda \sim N(0, 20)$; "HN" = unguarded study, $\lambda \sim \mathrm{HN}(0, 20)$. n = clean datasets in that study.

setting	study	n	CRPS			Brier q_{95}		
			i.i.d.	AR(2)	$\Delta\%$ (p)	i.i.d.	AR(2)	$\Delta\%$ (p)
4 strong	guarded	10	2.265	2.207	−2.6 (.161)	.0526	.0530	+0.8 (.539)
4 strong	HN	9	1.906	1.807	−5.2 (.049)	.0431	.0428	−0.7 (.213)
5 near-unit	guarded	8	2.695	1.851	**−31.3 (.039)**	.0623	.0552	−11.4 (.422)
5 near-unit	HN	8	2.780	1.795	**−35.4 (.012)**	.0468	.0395	−15.5 (.273)

uncertainty audit of the main simulation study, where station-level bootstrap and replicate-level Wilcoxon likewise found paired Brier differences undetectable.

3. **Why the Brier score is blind here.** At a q_{95} threshold the expected number of exceedance events per lead is $0.05\, n_s \approx 7$; over a 5-lead window the paired Brier difference is dominated by event-count noise, and its replicate-level variance swamps a 10–15% mean shift at $n = 8$–10. The CRPS integrates over all thresholds and does not pay this sparsity price.

3 Robustness to the ridge and recovery issues

The pathology and remedy ledgers (companion notes) support four invariance statements.

Table 2: Threat audit of the main result.

threat	evidence	verdict on Table 1
Reflected ridge (unprotected, would inflate scores of affected cells by up to 12× Brier)	two independent remedies deliver score-indistinguishable studies ($p = .52/.28$, $n = 35$); headline holds under both, $-31.3\% \to -35.4\%$	no effect once either remedy is active; protection is necessary (unguarded att-0 study would be contaminated)
τ-shape misrecovery under near-unit-root dependence ($\hat{a}$ CI covers truth 1/5 cells)	both methods fit the same cell's τ; defect is common to the pair	cancels in the paired difference
ϕ shrinkage in setting 4 (recovered 0.62 of 0.80)	weakens the AR(2) method's own tool where the gain is already undetectable	cannot fabricate the setting-5 gain; at most understates setting-4
Heavy-tail realizations ($\hat{a} < 2$, datasets s4 d1, s4 d6, s5 d5)	predictive clouds of all chains agree to $q_{99.9}$; scores within 1%	none; affects only the stability of the (now descriptive) spread statistic

The decision that follows, adopted for the thesis: no further model or prior adjustment beyond the already-adopted $\lambda \sim \mathrm{HN}(0, 20)$, and the residual issues are handled as reminder text.

4 Suggested thesis reminder text

The skew-t likelihood is flat along $(\lambda, \beta_0, \{z_t\}) \mapsto (\lambda^*, \beta_0^*, \{(m_t - \beta_0^*)/\lambda^*\})$, a ridge whose reflected branch ($\lambda^* < 0$) can capture a minority of MCMC chains on short training windows.

We remove the branch by the sign-anchored prior $\lambda \sim \mathrm{HN}(0, 20)$, which is justified because the sign of the skewness is known in simulation and identified by the sample skewness in applications; on the 40 positive-control fits this prior eliminated all sign reflections without measurable effect on any other parameter or on either forecast score. Two benign weak-identification effects remain and are reported rather than repaired. First, under near-unit-root dependence AR(2)$(0.15, 0.80)$ a 50-day window carries little information about the marginal of the latent variance process, so its shape parameter is over-estimated (posterior mean 3.9 against truth 3); the defect is shared by the compared methods and cancels in paired score comparisons. Second, occasional training windows realize heavy-tailed variance posteriors ($\hat{a} < 2$) for which moment-based predictive-spread statistics have no finite sampling variance; quantile-based spread summaries should be preferred when diagnosing such fits.

5 In flight

Experiment A (delivered to the compute host 31 July 2026) extends the design to settings 5 and 7 (ARFIMA $d = 0.45$ long memory, misspecified for any finite-order AR fit), all five blocks (seams $50, 80, 110, 140, 170$), methods i.i.d./AR(2)/AR(1), 10 datasets, under the HN-only policy with dataset-level pairing ($n = 10$; cell-level $n = 50$ as power sensitivity). It will answer the two open questions Table 1 cannot: whether the CRPS gain survives geometric-memory misspecification against hyperbolic-memory truth, and whether AR(2) separates from AR(1) rather than only from i.i.d.